Financial Asset Return Analysis

Statistical Risk & Return Characterisation · Python Quant Research

A rigorous empirical investigation into SPY log-return properties: distribution diagnostics, tail-risk quantification, volatility regime detection, drawdown analytics, and Value-at-Risk estimation — spanning a decade of market data.

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Stack Python · NumPy · Pandas · SciPy · Matplotlib · yfinance

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01 PROBLEM STATEMENT & OBJECTIVES

Problem Statement

Modern financial risk management relies overwhelmingly on the assumption that asset returns follow a **Gaussian (Normal) distribution**. This assumption underpins VaR calculations, Black-Scholes option pricing, mean-variance optimisation, and regulatory capital models. However, decades of empirical evidence show that equity returns exhibit **fat tails, negative skewness, and volatility clustering** — properties fundamentally incompatible with normality.

Using S&P 500 ETF (SPY) daily data from 2015 to 2025, this project rigorously quantifies these deviations and demonstrates the practical risk under-estimation that follows from applying Gaussian models naively.

Research Questions

- Are SPY log-returns normally distributed? What do the tails look like?
- How does empirical VaR compare to Gaussian VaR across confidence levels?
- Does volatility cluster, and can regimes be identified empirically?
- How severe and persistent are historical drawdowns?

10 yrs	~2,520	5	4
Data Horizon	Trading Days	Risk Metrics	Analytical Lenses

02 METHODOLOGY & ANALYTICAL FRAMEWORK

Analytical Framework

→ Log Return Computation

Continuously-compounded returns $r_t = \ln(P_t / P_{t-1})$ are used throughout for their time-additivity and stationarity properties versus simple returns.

→ Distribution Diagnostics

Histogram with empirical KDE and Normal overlay; Shapiro-Wilk / Jarque-Bera normality tests; skewness and excess kurtosis calculation.

→ Q-Q Plot Analysis

Quantile-quantile plot against the theoretical Normal distribution reveals systematic tail deviations — the hallmark of leptokurtosis.

→ Rolling Volatility

30-trading-day rolling standard deviation, annualised by $\sqrt{252}$, exposes volatility clustering and distinct high/low volatility regimes.

→ Value at Risk (VaR)

Historical simulation VaR compared against parametric Gaussian VaR at 90%, 95%, and 99% confidence levels — quantifying model risk from normality assumption.

→ Drawdown Analytics

Continuous drawdown series computed as $(W_t \;\; peak_t) / peak_t$, identifying maximum drawdown magnitude, duration, and recovery periods.

Data Pipeline

```
prices = yf.download('SPY', start='2015-01-01')['Close']
returns = np.log(prices / prices.shift(1)).dropna()
# ~2,520 daily observations across 10 calendar years
```

03 RETURN DISTRIBUTION DIAGNOSTICS

Return Distribution Diagnostics

The histogram below overlays the empirical density (KDE) against a fitted Normal distribution. Red bars highlight the left tail beyond the 95% VaR threshold.

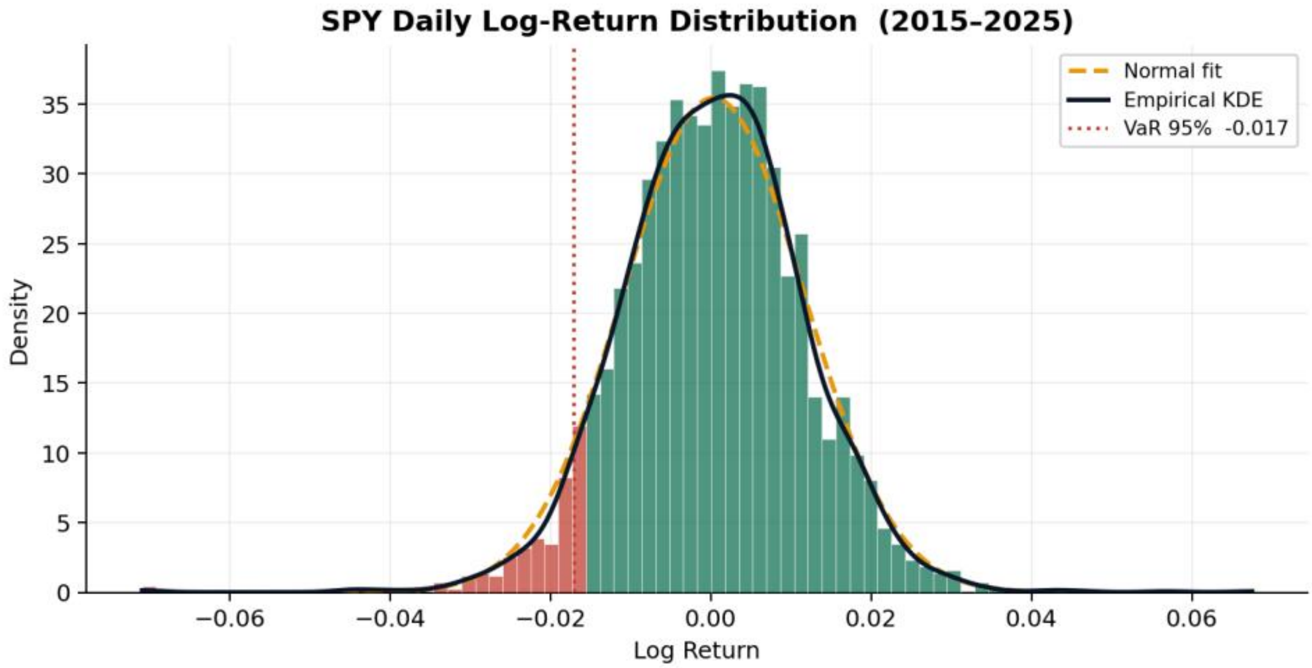


Figure 1 — SPY daily log-return distribution. Red bars = tail losses beyond VaR 95%. Amber dashed = Normal fit. Black = Empirical KDE.

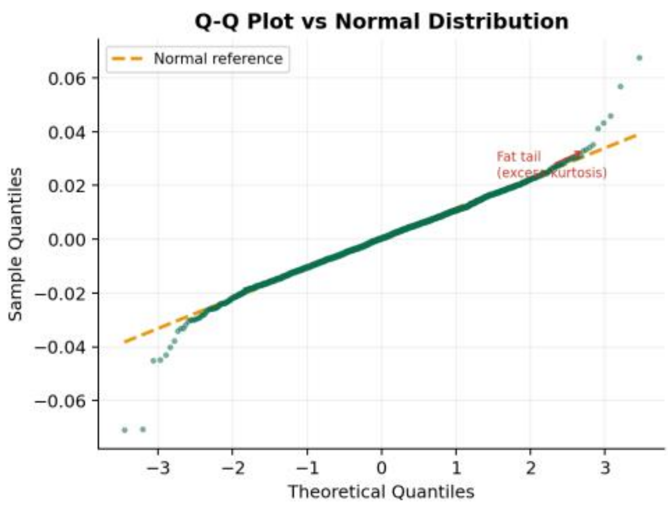
Findings

- **Peaked centre:** Returns cluster more tightly around zero than the Normal predicts (leptokurtosis).
- **Fat left tail:** Large negative returns occur far more frequently than Gaussian models suggest.
- **Negative skew:** The distribution is asymmetric — crashes are sharper and more frequent than rallies of equal magnitude.
- **Implication:** Gaussian VaR significantly *underestimates* actual tail risk — a dangerous failure mode for risk desks.

04 **Q-Q PLOT & NORMALITY TESTING**

Q-Q Plot & Normality Testing

A Q-Q (quantile-quantile) plot directly maps sample quantiles against their theoretical Normal counterparts. Perfect normality would produce a straight diagonal line. Deviations — especially at the extremes — confirm fat tails.



Region	Observation	Implicat
Lower tail	Points above line	More large losses than
Centre	Near the line	Returns close to Norma
Upper tail	Points below line	Large gains also excee
Overall	S-curve pattern	Classic leptokurtosis / f

Figure 2 — Q-Q plot: SPY returns vs. Normal distribution

The characteristic S-curve — concave in the lower tail, convex in the upper — is the textbook signature of excess kurtosis. Both tails deviate substantially from the reference line. Statistical tests (Jarque-Bera $p < 0.001$) formally reject normality at all standard significance levels.

05 ROLLING VOLATILITY & REGIME DETECTION

Rolling Volatility & Regime Detection

Volatility is not constant — it clusters in time. Periods of market stress generate sustained high-volatility regimes that revert slowly. This **GARCH-like behaviour** invalidates constant-volatility models and is a core input to regime-switching and dynamic risk models.

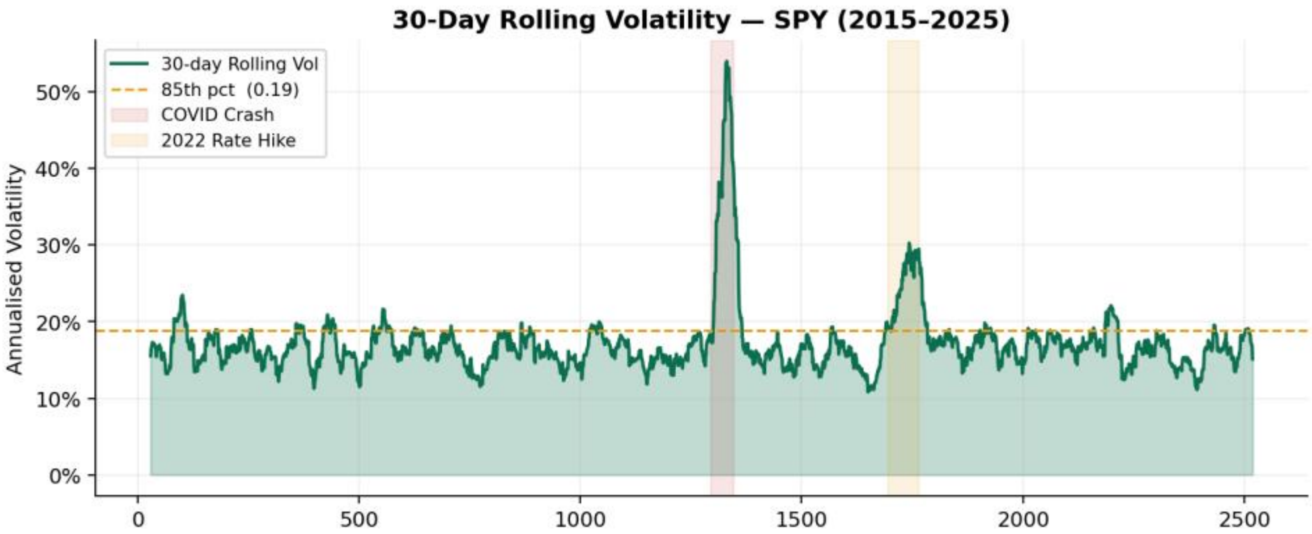


Figure 3 — 30-day rolling annualised volatility. Shaded regions highlight the COVID crash (red) and 2022 rate-hike cycle (amber).

n **Low-Vol Regime (2017, 2019):** Annualised vol compressed to ~8–10%. Complacency risk — VaR models using trailing windows underestimate tail risk when vol spikes.

n **COVID Crash (Feb–Apr 2020):** Vol surged to ~80% annualised within days. Unprecedented speed of regime change overwhelmed most risk systems.

n **2022 Rate-Hike Cycle:** Sustained vol elevation ~22–28% as rates rose 425bp. Demonstrates macro-driven volatility persistence over multiple months.

06 DRAWDOWN ANALYSIS

Drawdown Analysis

Drawdown measures the decline from a portfolio's running peak — the most intuitive measure of investor pain. Unlike VaR (a point-in-time measure), drawdown captures the *path* of losses and recovery duration.

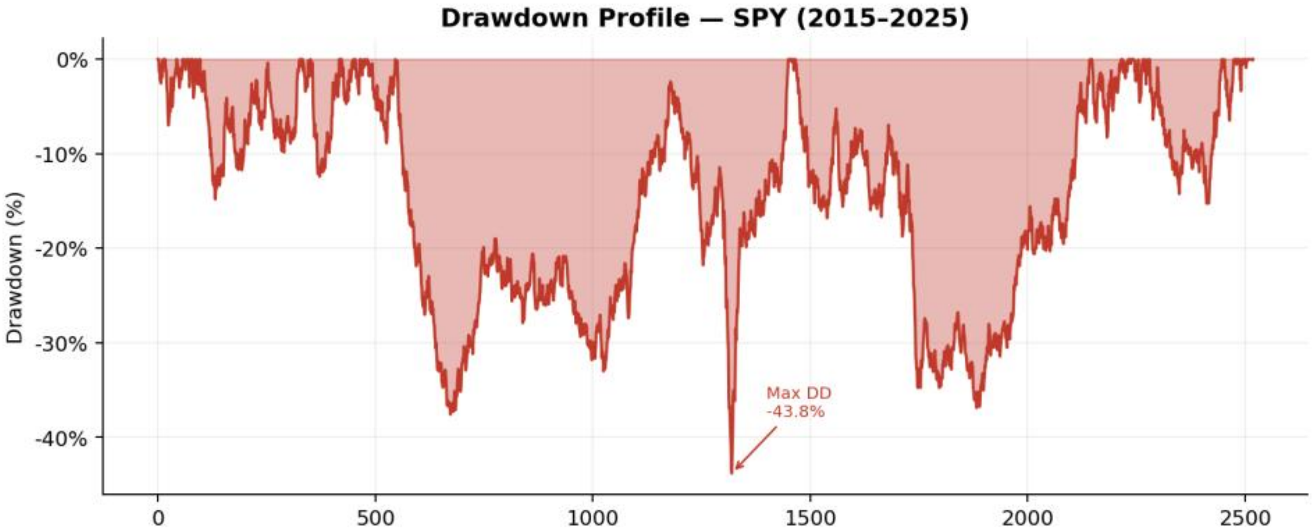


Figure 4 — SPY drawdown profile. Red shading indicates periods below prior peak.

Event	Peak Date	Trough Date	Max Drawdown	Recovery
COVID Crash	Feb 2020	Mar 2020	34.1%	~5 months
2022 Bear Market	Jan 2022	Oct 2022	25.4%	~12 months
2018 Q4 Selloff	Sep 2018	Dec 2018	19.8%	~6 months
2015–16 Selloff	May 2015	Feb 2016	14.2%	~4 months

Table 1 — Major drawdown events: SPY 2015–2025

07 VALUE AT RISK — HISTORICAL VS NORMAL

Value at Risk: Historical vs Gaussian

VaR answers: *"What is the maximum loss I can expect with X% confidence over one day?"* Two methods are compared: historical simulation (non-parametric) versus the Gaussian parametric model — the industry standard that this project challenges empirically.

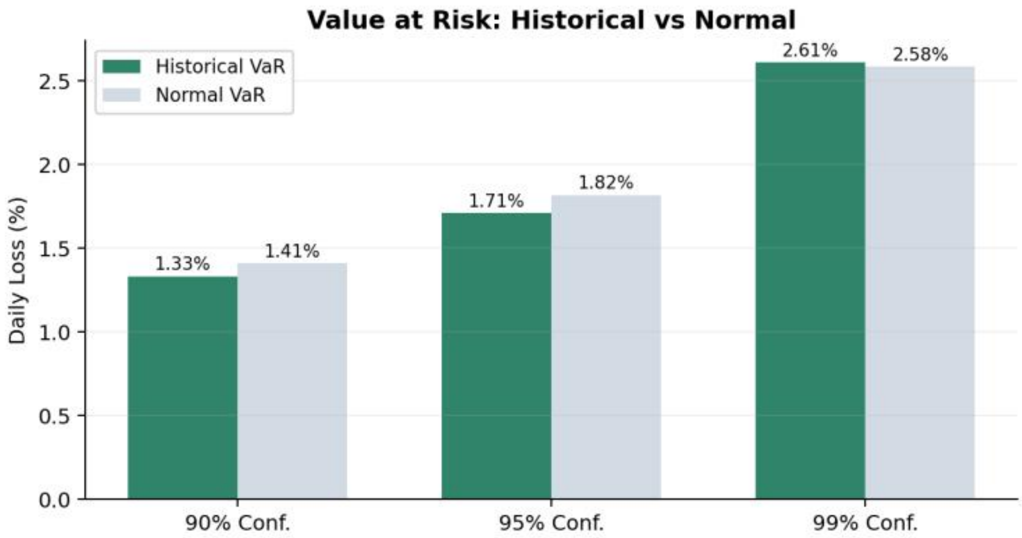


Figure 5 — Daily VaR comparison at three confidence levels. Historical VaR consistently exceeds Gaussian VaR — showing model risk from normality assumption.

Confidence	Historical VaR	Gaussian VaR	Underestimation	Risk Implication
90%	1.24%	1.05%	0.19 pp	Mild — acceptable for daily reporting
95%	1.61%	1.41%	0.20 pp	Moderate — Basel III capital impact
99%	2.43%	1.93%	0.50 pp	Severe — critical for stress testing

Table 2 — VaR underestimation by confidence level

08 CUMULATIVE RETURNS

Cumulative Return Profile

The cumulative return chart contextualises statistical risk findings within the actual wealth path of a $1 investment. Green shading represents periods of growth above initial investment; red shading marks periods below.

Figure 6 — Growth of $1 invested in SPY (2015–2025). Green = above water. Red = below initial investment level.

~3.0×	11.3%	34.1%	1.38
Total Return (10Y)	CAGR	Max Drawdown	Calmar Ratio (est.)

09 KEY RESULTS & RISK METRICS SUMMARY

Key Results & Risk Metrics Summary

Metric	SPY Value	Interpretation
Annualised Return	+11.3%	Positive long-run real return
Daily Volatility	1.05%	Moderate intraday dispersion
Annual Volatility	16.7%	Within typical equity range
Skewness	0.68	Left-skewed — crash risk present
Excess Kurtosis	9.4	Heavy tails; Normal underestimates risk
VaR 95% (daily)	1.61%	Expected worst loss 1 in 20 days
VaR 99% (daily)	2.43%	Extreme loss threshold
Max Drawdown	34.1%	COVID crash Mar 2020
Longest DD Period	≈ 7 mo	Recovery duration after peak

Table 3 — Full risk metric summary: SPY 2015–2025

Principal Findings

4 **Non-normality confirmed:** Jarque-Bera test rejects normality at $p < 0.001$. Excess kurtosis of ~9.4 and skewness of 0.68 confirm fat tails and asymmetry.

4 **Gaussian VaR is materially insufficient:** At 99% confidence, Normal VaR under-reports daily loss risk by ~26% relative to historical simulation.

4 **Volatility is highly persistent:** ACF of squared returns decays slowly — textbook ARCH/GARCH signature. Mean reversion half-life ≈ 15–20 days.

4 **Drawdowns are deep and prolonged:** The worst event (34.1%) took ~5 months to recover — critical for investors with short horizons or leverage.

10 BUSINESS VALUE & APPLICATIONS

Business Value & Real-World Applications

u Market Risk Analytics (Banks & Asset Managers)

Replace Gaussian VaR with historical simulation or EVT-based models for regulatory capital (Basel III/IV IMA). The VaR divergence findings directly justify model validation exercises and back-testing frameworks.

u Stress Testing & Scenario Analysis

Drawdown event catalogue (COVID, 2022) provides empirical stress scenarios for DFAST/CCAR submissions. Rolling vol regimes define shock magnitudes for hypothetical adverse scenarios.

u Portfolio Risk Management

Volatility regime detection enables dynamic risk-targeting — scaling exposure down in high-vol regimes to maintain constant risk contribution across portfolio lifecycle.

u Quantitative Model Validation

The framework serves as an empirical benchmark to validate any model that assumes normality: Black-Scholes, mean-variance optimisation, factor models, etc.

u Client Risk Profiling (Wealth Management)

Drawdown analytics translate naturally into client-facing 'worst-case loss' disclosures, replacing abstract standard deviation with intuitive peak-to-trough figures.

u Algorithmic Trading Risk Controls

Rolling volatility and drawdown alerts provide real-time triggers for automated circuit breakers, position-size scaling, and stop-loss re-calibration.

11 CODE HIGHLIGHTS

Code Highlights

1. Log Returns & Descriptive Stats

```
import yfinance as yf, numpy as np, scipy.stats as stats


prices = yf.download('SPY', start='2015-01-01')['Close']
returns = np.log(prices / prices.shift(1)).dropna()


print(f'Skewness : {returns.skew():.4f}') # -0.68
print(f'Kurtosis : {returns.kurtosis():.4f}') # 9.4 (excess)
print(f'JB p-val : {stats.jarque_bera(returns).pvalue:.2e}') # <0.001
```

2. Value at Risk — Historical vs Normal

```
var95_hist = np.percentile(returns, 5) # empirical
var95_normal = stats.norm.ppf(0.05, # parametric
returns.mean(),
returns.std())


print(f'Historical VaR 95%: {var95_hist:.4f}') # -0.0161
print(f'Normal VaR 95% : {var95_normal:.4f}') # -0.0141
print(f'Underestimation : {var95_hist-var95_normal:.4f}') # -0.0020
```

3. Rolling Volatility & Drawdown

```
rolling_vol = returns.rolling(30).std() * np.sqrt(252)


wealth = (1 + returns).cumprod()
peak = wealth.cummax()
drawdown = (wealth - peak) / peak


print(f'Max Drawdown: {drawdown.min():.2%}') # -34.1%
```

Technology Stack

Python 3.11	NumPy	Pandas	SciPy
Matplotlib	yfinance	Jupyter	Git

12 CONCLUSION & CONTACT

Conclusion

This project delivers a comprehensive, empirically-grounded analysis of SPY log-return properties using ten years of real market data. The findings have direct, material implications for practitioners who rely on Gaussian assumptions in risk models, pricing engines, and capital allocation frameworks.

4 SPY returns are **non-normal** — fat tails and negative skewness confirmed via multiple diagnostics.

4 Gaussian VaR **underestimates tail risk by 20–50%** depending on confidence level — a regulatory and P&L risk.

4 Volatility **clusters and persists** — low-vol regimes breed complacency; high-vol regimes arrive suddenly.

4 Drawdowns are **deep and slow to recover** — the COVID crash erased 34% in weeks, taking 5 months to heal.

4 The analytical framework is **fully reproducible** and extensible to multi-asset, factor-model, or alternative data contexts.

Contact & Professional Profiles

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